

PRATIK BYATHNAL

- Phone: +1 437-998-8735
- Email: pratikbyathnal@gmail.com
- Location: Huron Street, London, Ontario, Canada - N5Y 4K9
- LinkedIn: <https://linkedin.com/in/pratik-byathnal>

Summary

Tailored for Android OS Engineer opportunities at Infotainment @ Rivian and Volkswagen Group Technologies. Focus: Kernel and embedded Linux development, Yocto image build and board bring-up, Debugging and performance workflows. Experienced Software Engineer with industry experience in Android development, fintech systems, and systems-oriented engineering across Epifi Technologies and Samsung Research Institute Bangalore. Strong background in building production-scale Android applications using Kotlin and Java, with hands-on experience developing investment and financial workflow features for consumer fintech platforms. Currently pursuing a Master of Computer Science at Western University with research interests in cybersecurity, distributed systems, parallel computing, and low-level systems engineering. Experienced in mobile architecture, asynchronous programming, automated testing, API integration, and performance optimization. Academic and personal research exposure includes Android platform security, Secure Boot and Verified Boot mechanisms, embedded systems, and distributed mobile computing.

Skills

- **Tags:** android, frontend, linux_kernel, cyber_security, general
- **Tags:** linux_kernel, system_security, cyber_security, general
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- **Mobile & Platform Engineering:** Android SDK, Jetpack (MVVM, Room, Navigation, LiveData), Coroutines, Firebase, REST APIs, gRPC, Android Architecture Components, AOSP Fundamentals, Embedded Linux Fundamentals

Experience

Software Engineer — Samsung Research Institute (SRI-B), Bangalore, India | Jul 2021 – Jan 2022

- Developed and optimized 5G NR MAC Layer features aligned with 3GPP specifications for next-generation wireless communication systems and modem software stacks.
- Worked on beamforming, uplink/downlink scheduling, and timing-related workflows within Linux-based telecom development environments.
- Implemented beam ID mapping mechanisms linking narrow beams to corresponding wide beam groups in 5G communication systems, reducing signaling overhead and improving beam selection and handover efficiency.

Android Developer — Epifi Technologies Pvt Ltd (Fi.Money), Bangalore, India | Sept 2022 – Nov 2023

- Developed and shipped Android fintech features for a platform serving approximately 700K–2.1M active users and 10M+ Google Play Store downloads.
- Contributed to Mutual Funds and US Stocks investment modules by building onboarding workflows, portfolio dashboards, and interactive stock visualization interfaces.
- Designed and implemented scalable Android architecture using MVVM, Jetpack

Components, Room Database, View Binding, Dagger, and Kotlin Coroutines, improving code maintainability and lifecycle-aware state handling.

Projects

Embedded Linux Image & Kernel Work — Raspberry Pi 3

- Built and maintained Yocto Project layers and BitBake recipes to produce custom rootfs and images targeting Raspberry Pi 3 (armv7). Managed BSP packages, machine configuration, and image recipes for tight disk and memory constraints.
- Configured and customized Linux kernel builds (e.g., 5.x series), edited Raspberry Pi 3 Device Tree Source (DTS) files to enable board peripherals, and developed kernel patches and modules for board-specific drivers.
- Set up cross-compilation toolchains (arm-linux-gnueabi), reproducible image-build pipelines, and automated image generation and testing in CI; used QEMU for early functional smoke tests and flashed images to Raspberry Pi 3 hardware for validation.

Android SMS Parser — Epifi Technologies Pvt Ltd

- Designed and implemented a privacy-conscious SMS parsing system within a fintech Android application to identify and track telecom promotion messages from configurable provider channels without accessing sensitive user message content.
- Built scheduled background workflows using Android JobScheduler to periodically scan for server-configured telecom sender patterns such as branded operator channels and campaign identifiers.
- Developed secure filtering and parsing logic to process anonymized sender metadata for telemetry and market analysis workflows while maintaining user consent and data privacy standards.

Education

- Masters of Science in Computer Science, Western University, London, Ontario | 2025 – 2026
- Focus: Research in cybersecurity and distributed/parallel systems
- Graduate Certificate in Mobile Application Development (Co-op), College in London, Ontario | 2024 – 2025
- GPA: 4.1/4.2
- Focus: Progressive Web Apps, Android, iOS, React
- Bachelor of Technology in Computer Science and Engineering, PES University, Bangalore, India | 2017 – 2021
- CGPA: 7.9/10
- Focus: System and Core Computing, Big Data, Cloud, ML, AI